

Shahaddin Gafarov

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EDUCATION

University of Florida

Doctor of Philosophy in Computer Science

Gainesville, FL

2025 – Present

University of South Florida

Bachelor of Science in Computer Science

Tampa, FL

2021 – 2024

EXPERIENCE

Graduate Research Assistant - Bioinformatics and Medical AI

Apr 2025 – Present

University of Florida

Gainesville, FL

- Worked on a multimodal deep learning framework on cancer survival prediction.
- Developing a robustness benchmark for evaluating contemporary machine learning techniques on biological data.
- Lead Teaching Assistant for Advanced Data Structures by Distinguished Professor Dr. Sartaj Sahni.

Undergraduate Research Assistant

Dec 2023 – Dec 2024

University of South Florida

Tampa, FL

- Defined robotic systems for benchmarking and developing multi-object grasping solutions.
- Detected 6D poses of top-layer objects in the target using deep learning techniques.

Full-Stack Developer

Apr 2022 – Aug 2022

MusicLessonHub

Boise, ID (Remote)

- Contributed to a startup connecting students and teachers across the United States.
- Collaborated with designers to develop interfaces using front and back-end tools.

Software Engineer, Databases

Jun 2020 – Jun 2021

AzeriMed LLC

Baku, Azerbaijan

- Worked on aptekonline.az database, Azerbaijan's first online pharmaceutical store for business purposes.
- Collaborated with developers on database error handling, backend reliability, and data-related issue resolution.

PROJECTS AND COLLABORATIONS

Bioinformatics and Medical AI

- **SPACT**: Novel multimodal network for predicting cancer survival based on tissue using WSI + Multi-Omics.
- **PLATO**: Dynamic graph-based learning model, focusing on predicting future biological interactions such as drug-drug, gene-gene, and protein-protein.

Programming

- **Delphi**: Implemented the grammar and parser/lexer/interpreter from scratch in Java for the Delphi Language.
- **InCollege**: Built a student-focused networking platform with user search and profiles using Agile method.

PUBLICATIONS

Öğülmüş, F. E., **Gafarov, S.**, Almaloğlu, Y., Özdemir, B. H., Ok Atılğan, A., Demir, D., Özen, Ö., Keleş, G. E., Kahveci, T., & Turan, M. (2026). *SPACT: A Clustering-Driven Multi-Modal Framework for Survival Prediction Using Genomic and Histopathology Data*. *Medical Image Analysis*, 112, 104078.

Khatib, T., **Gafarov, S.**, Strikowski, J. D., Turan, M., & Kahveci, T. (2025). *PLATO: Predicting Longitudinally-Aligned Time Observations of Biological Networks*. In *Proc. of the 16th ACM Int'l Conf. on Bioinformatics, Computational Biology and Health Informatics (BCB '25)*.

Chen, T., Frumento, R., Pagnanelli, G., Cei, G., Keth, V., **Gafarov, S.**, Gong, J., Ye, Z., Baracca, M., D'Avella, S., Bianchi, M., & Sun, Y. (2025). *Benchmarking Multi-Object Grasping*. *IEEE Robotics and Automation Letters*, 10, 9510–9517.

TECHNICAL SKILLS

Languages: Python, C/C++, C#, SQL (MySQL), Java, JavaScript, HTML/CSS

Libraries: PyTorch, NumPy, Pandas, Matplotlib, TensorFlow

NOTEWORTHY ACHIEVEMENTS AND AWARDS

University of Florida: Full tuition waiver and stipend, PhD Program

University of South Florida: Green & Gold Presidential Award

National Informatics Olympiad of Azerbaijan: Bronze Medal